SESSION 20

IPL Match Data Analysis

Q-1: Download a sample IPL match data CSV and load it into a new SQL table

```
CREATE TABLE ipl_matches (
    match_id INT PRIMARY KEY,
    team1 VARCHAR(50),
    team2 VARCHAR(50),
    winner VARCHAR(50),
    venue VARCHAR(100),
    match_date DATE
);
```

Q-2: SQL query to select all matches where *Mumbai Indians* played, then export results

```
SELECT *
FROM ipl_matches
WHERE team1 = 'Mumbai Indians' OR team2 = 'Mumbai Indians';
```

Q-3: Connect Microsoft Excel to your SQL database → Import *ipl_matches* table → Create a pivot table

- Rows: Team names (team1 and team2 combined).
- Values: Count of matches played. This shows the total matches played by each team.

Q-4: Connect Power BI Desktop to your SQL database → Import *ipl_matches* table → Create a pivot chart

- Axis: Team names (winner column).
- Values: Count of wins. This chart shows the number of wins per team.

Q-5: Use Python with pandas and sqlalchemy to filter Mumbai Indians wins and insert into *mi_wins* table

```
import pandas as pd
from sqlalchemy import create_engine

# Read the exported CSV
df = pd.read_csv("mi_matches.csv")

# Filter matches where Mumbai Indians won
mi_wins = df[df['winner'] == 'Mumbai Indians']

# Connect to SQL database
engine = create_engine("mysql+pymysql://username:password@localhost/dbname")

# Insert filtered data into new table
mi_wins.to_sql("mi_wins", con=engine, if_exists="replace", index=False)
```